

Management Accounting of Outsourced Manufacturing along the Global Value Chain

Project Study in the Specialization Finance & Accounting

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1. Introduction

1.1 Background and Rationale

In today's paced business world there is an increasing focus on outsourcing and subcontracting. Both practitioners and academia are constantly seeking to understand and delve deeper into these processes. The purpose of this paper is to contribute to the expanding knowledge about this aspect of corporate operations. We will achieve this by conducting an analysis of the intricacies within the context of reporting and accountability.

The terms outsourcing and subcontracting are often used interchangeably in today's industry. This questionably straightforward approach has the potential to cause confusion, however, as they are inherently distinct processes. Although both can be considered similar in nature, it is essential to distinguish between them with different core objectives, namely with outsourcing primarily focuses on cost reduction, whereas subcontracting emphasizes creating or adding value. In addition, firms often opt for subcontracting when they lack specific capabilities or resources, not just for cost-cutting.

1.2 Objectives of the Study

Outsourcing and subcontracting receive, without doubt a significant amount of attention from the business world. Despite this fact, the truth has been that there has only existed a limited exploration of these processes with all their aspects taken into consideration, namely the many subcategories. Subcontracting is an incredibly diverse phenomenon that consists of several varieties, which all warrant an independent exploration into their intricacies, this paper, however, will focus solely on the subcategory of Buy / Sell subcontracting. Two parties of particular interest in this subcontracting scenario are the OEM (Original Equipment Manufacturer) and the SP (Service Provider). A quick rundown of the process would be as follows: The process is set off by the OEM selling the components that they have previously procured to the SP, who subsequently perform a task or service. After an agreed upon period, the OEM purchases the finished goods that are produced from the components sold initially by the OEM. Safe to say, this unique and detailed process demands a hand-tailored set of accounting requirements that diverge significantly from the standard and particularly affect the OBS (Off-Balance Sheet) section of financial statements.

1.3 Scope and Limitations

The scope of this paper lies within the boundaries of subcontracting and focuses specifically on a subcategory thereof: Buy / Sell Subcontracting. With the help of financial regulations established by the International Financial Reporting Standards (IFRS) concerning the purchase obligations that arise from Buy/Sell Subcontracting, this paper's main objective of ensuring reliable and truthful disclosure of information related to the process is imperative in order to provide investors with insights into contractual obligations that have the potential to affect future liquidity and, subsequently, the financial stability of these companies. It is thus vitally important to examine any commitments to purchase, which significantly affects liquidity, among many other aspects of Buy / Sell subcontracting.

2. Subcontracting Fundamentals

2.1 Definition of Subcontracting

Subcontracting is a business process where a company engages the services of an external individual or firm to carry out a specialized task or function that the internal team may not have the capability to perform. However, it does not involve allocating entire departments or jobs on a permanent basis.¹ It is just a contractual agreement between the two parties for a limited time period. Also, this strategic decision allows a company to focus on its core competencies by allocating specific tasks to specialized external partners. With this allocation, companies can leverage their technological skills to their fullest, mitigate the risk associated with unfamiliar or highly specialized tasks and, of course, gain a competitive advantage over other companies.

The subcontracting process involves close collaboration with another business to perform tasks that are usually outside the boundaries of a firm's core strength. Under normal circumstances, the hiring party would retain a reasonable degree of control over the process up until completion. Opening to opportunities to collaborate with firms that have competencies that complement a firm's own allows both to utilize their skills to their fullest extent in an arrangement where external experts with the required skill sets are relied upon. An example of this process would be as follows: Let us consider for a moment a manufacturing plant producing complex machinery systems, which despite being an expert in manufacturing, lacks the necessary expertise in developing the control systems for the

¹ (Difference between Outsourcing and Subcontracting, kein Datum)

machinery produced. This is a prime opportunity for subcontracting the production of the control systems to an external entity that has the experience in manufacturing the component. Through collaboration, an enhanced product is manufactured, and a more comprehensive and diverse set of skills is utilized. Ultimately, subcontracting is regarded as a strategic tool for accessing external competencies to deliver projects or achieve tasks that lie beyond the firm's core strengths.

2.2 Distinguishing Between Subcontracting and Outsourcing

Both outsourcing and subcontracting have benefits and risks for the companies in the long run. There are several similarities between the two as well. However, it is important to understand that they serve distinct purposes and have varying degrees of involvement from external parties.

2.2.1 Definition

Outsourcing and subcontracting, though related, differ fundamentally in their definitions. **Outsourcing** is mainly a cost-cutting strategy where a company permanently hands over all business functions within their firm to external third parties.² Usually, this cost-saving measure involves a focus on reducing labor expenses. Many companies choose to outsource functions that they could handle internally. They do this to save costs and often seek partnerships. On the other hand, subcontracting involves hiring individuals or external companies to complete tasks that cannot be done within the organization itself.

2.2.2 Duration

When it comes to **outsourcing**, one of the objectives is to establish a permanent relationship with external service providers. The idea is that once a business function or process is outsourced, the provider will manage it long-term. This long-term commitment aims to optimize cost-effectiveness, and operational efficiency and provide support for core functions. On the other hand, **subcontracting** is typically done on a basis. It usually involves project tasks or assignments for a defined period. This temporary nature offers companies flexibility. Allows for responses to project requirements without the need for a long-lasting commitment like outsourcing. Therefore, deciding between outsourcing and subcontracting often depends on whether a company requires permanence or adaptability in addressing its project-related needs.

² (THE INVESTOPEDIA TEAM, 2021)

2.2.3 Involvement Levels

In the case of **outsourcing**, the service provider usually initiates the formation of a contract. Nevertheless, such a service provider usually does not have much control over the job, although a contract is involved. The hiring company has some say over its operations but assigns most operational authority to the external specialist. However, **subcontracting** grants the employer a degree of authority. In a subcontracting arrangement, the service provider possesses the ability to negotiate, create, and outline all elements of the product. Engaging a company guarantees oversight of the process, adaptability, and a hands-on approach to hiring procedures.

2.3 Motivations and Benefits of Subcontracting

Subcontracting is a used business strategy across various industries, and its popularity is consistently on the rise. There are factors that contribute to this trend as organizations recognize the potential advantages and motivations associated with adopting subcontracting practices.

One primary factor is the desire to improve efficiency. Many organizations frequently face challenges in terms of lacking skills or resources. Developing these capabilities in-house can be quite challenging, which is why outsourcing provides an efficient solution. By assigning tasks or services to entities, especially those with proven expertise, businesses can ensure precise execution, speed and a high level of quality that might be difficult to achieve internally.

Cost reduction is another aspect of subcontracting for companies. Collaborating with subcontractors, particularly those located in the cost region's economies of scale can be achieved. This allows businesses to easily adapt their operations based on fluctuating demand while maintaining stability.

Subcontracting arrangements often provide a flexible cost structure that allows companies to adapt financially without making fixed investments. The ability to adapt quickly in today's changing business world is highly valuable and subcontracting offers just that. Companies can swiftly respond

to market fluctuations by expanding their operations during periods or scaling back during quieter times.

Moreover, subcontracting goes beyond transactions; it fosters innovation. When companies collaborate with subcontractors, they gain access to a wealth of knowledge and experience beyond obtaining services. These collaborations often ignite ideas and stimulate the development of innovative solutions and products that might not have been possible within a single organization.

In summary, while the financial advantages of subcontracting, such as cost savings, are certainly appealing, there are also reasons to consider this strategy. These include increased efficiency, strategic positioning, and fostering innovation. Companies that wisely recognize and leverage these benefits are well-equipped to navigate today's ever-changing business landscape, ensuring both sustainability and growth in our evolving world.

2.4 Various Types of Subcontracting Scenarios

2.4.1 Buy/Provide

The subcontracting buy/provide model stands out as a strategy among subcontracting strategies. In this model, the main entity, usually referred to as the OEM (Original Equipment Manufacturer), purchases raw materials or components and then supplies them to a subcontractor. The subsidiary manufactures or assembles these materials for the OEM. This method enables the equipment manufacturer (OEM) to retain authority over the excellence and specifications of components guaranteeing that the end product satisfies its desired benchmarks. Additionally, it takes advantage of the subcontractor's expertise in manufacturing or assembling, leading to cost optimization and improved production efficiency.³

2.4.2 Directed Buy

The Directed Buy subcontracting model is the one of the types of subcontracting wherein the OEM, picks a particular subcontractor for a task, and mandates the suppliers from whom the subcontractor

³ (Team TranZact, 2023)

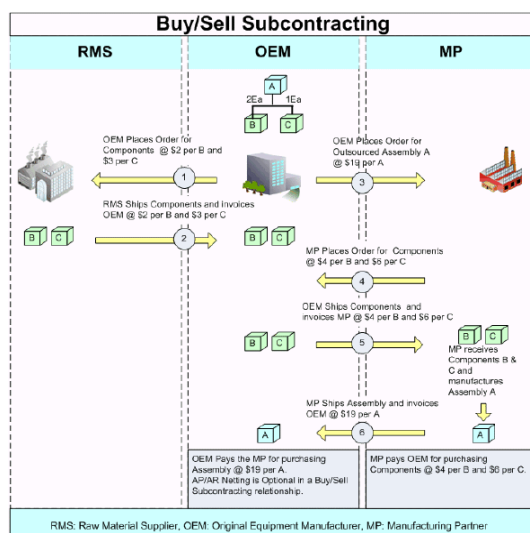
should procure components or materials. This approach helps the OEM to exert better control over the supply chain, guaranteeing specific quality standards, pricing, or tactical relationships are upheld. By directing the selection of both the subcontractor and the suppliers, the OEM aims to achieve a balance between cost-efficiency and quality assurance, as well as aligning the production process with its overall strategic objectives.⁴

2.4.3 Turnkey

A highly streamlined subcontracting model, turnkey subcontracting mandates the SP to take over the manufacturing wholly, from its inception to completion. In this setup, the subcontractor provides a so-called "turnkey" solution, in which they handle all aspects of the project, including design, procurement, construction, installation, and testing. Upon completion, control over the product is effectively transferred to the OEM. Turnkey subcontracting differs itself from other instances of subcontracting over its convenience, which the OEM relinquishes.⁵

2.4.4 Buy/Sell

In the Buy/ subcontracting model there is a way of procuring and subcontracting where the original equipment manufacturer (OEM) sells components that have already been acquired to a dedicated service provider or subcontractor. Then SP performs services on these components by either making modifications or enhancements. Afterward the OEM repurchases the finished product. This model brings advantages for the OEM in terms of accounting and operations. It allows them to leverage the subcontractors' capabilities while still maintaining control over aspects of the production process.⁶



⁴ (Team TranZact, 2023)

⁵ (Team TranZact, 2023)

⁶ (Overview of Oracle Subcontracting)

Image-1: Buy/Sell Subcontracting Process⁷

3. Accounting, Financial Implications, and Regulations

3.1 International Financial Reporting Standards (IFRS 15 & IFRS 7)

The IFRS (International Financial Reporting Standard) provides the set of rules and regulations that apply to accounting for Buy / Sell subcontracting, effectively implementing these however requires a fundamental understanding of both the standards themselves as well as how they relate to our specific case of subcontracting. Unlike Japan or Taiwan, IFRS does not especially relate to Buy / Sell subcontracting (whereas the Japanese Subcontract Act explicitly focuses on this instance, although with slight differences and called chargeable subcontracting), which is why the IFRS standards considered are somewhat diverse.

Accordingly, the most relevant books that come into play are the IFRS 15 & the IFRS 7, with topics of “Revenue from Contracts with Customers” and “Financial Instruments: Disclosures” respectively. Both sources are instrumental in setting the standards that must be applied to Buy / Sell subcontracting, as they illustrate, in effect how the process translates into financial statements.

IFRS 15

Published May 2014, IFRS 15 is invaluable as a regulatory guide to contracts with customers and revenue generated from performance obligations. With the objective to “ (...) establish the principles that an entity shall apply to report useful information to users of financial statements about the nature, amount, timing and uncertainty of revenue and cash flows arising from a contract with a customer.”⁸ IFRS 15 becomes more often than not the main influence for the accounting principles that apply to Buy / Sell subcontracting. It states that this standard is to be applied to all contracts with customers with the exception of some cases such as lease agreements as well as insurance contracts⁹, and defines a customer as “ (...) a party that has contracted with an entity to obtain goods or services that are an

⁷ (Overview of Oracle Subcontracting)

⁸ (IFRS Foundation, 2015, S. para. 1)

⁹ (IFRS Foundation, 2015, S. para. 5)

output of the entity's ordinary activities in exchange for consideration.”¹⁰ Furthermore, it is worth noting that according to this standard “A counterparty to the contract would not be a customer if, for example, the counterparty has contracted with the entity to participate in an activity or process in which the parties to the contract share in the risks and benefits that result from the activity or process (such as developing an asset in a collaboration arrangement) rather than to obtain the output of the entity's ordinary activities.”¹¹

Furthermore, the IFRS 15 introduced a revenue recognition model consisting of five steps. According to this model revenue can only be recognized once the contracts have been identified performance obligations have been determined transaction prices have been established and the allocation of transaction prices to performance obligations has been made. Only after these steps can revenue be officially recognized. The aim of IFRS 15 is to simplify the process of revenue recognition by promoting an model that relies on factual information regarding the contract.

There is a much greater number of relevant paragraphs for Buy / Sell subcontracting than illustrated here in IFRS 15 which will be delved much deeper into in the coming chapters. Some that are especially of interest are paragraphs 33 & 34, which discuss the transfer of ownership of assets in addition to how an underlying repurchase agreement would affect said transfer and paragraphs 22, 23 & 26, where the key component of performance obligations come into play. These paragraphs define what a performance obligation is and elaborate on the concept of distinctness of goods and services provided.

IFRS 7

Whereas IFRS 15 highlights contractual reporting obligations, IFRS 7 has the objective “ (...) to require entities to provide disclosures in their financial statements that enable users to evaluate: (a) the significance of financial instruments for the entity's financial position and performance; and (b) the nature and extent of risks arising from financial instruments to which the entity is exposed during the period and at the end of the reporting period, and how the entity manages those risks.”¹².

¹⁰ (IFRS Foundation, 2015, S. para. 6)

¹¹ (IFRS Foundation, 2015, S. para. 6)

¹² (IFRS Foundation, S. para. 1)

It applies to a broad scope of financial instruments, with paragraph 5 stating that “This IFRS applies to contracts to buy or sell a non-financial item that are within the scope of IFRS 9.”¹³, which is extremely relevant for Buy / Sell subcontracting as the process fundamentally involves an obligation for the OEM to purchase the finished good.

IFRS 7 also handles several topics of interest for the case of Buy / Sell subcontracting, such as the liquidity risk inherent in the transactions as well as how it should be managed and disclosed by stating that “An entity shall disclose: (a) a maturity analysis for non-derivative financial liabilities (including issued financial guarantee contracts) that shows the remaining contractual maturities.

(...) (c) a description of how it manages the liquidity risk inherent in (a) and (b).”¹⁴ The liquidity risk is an aspect of Buy / Sell subcontracting that the Japanese Subcontract Act aims to regulate by timing cash transactions in a way that creates less of a burden for the MP; in contrast, the IFRS refrains from making a specific statement for subcontracting. Regardless, the obligation to disclose contractual maturities in a way that illustrates liquidity risk transparently is essential for allowing truthfulness in a financial reporting context.

Even though our case of subcontracting is not a financing arrangement, obligations and transactions that arise from it are disclosed in accordance with IFRS 7. This includes off-balance sheet liabilities as well as many other nuances of the accounting for this process. These items are essential for a complete disclosure, as they outline the benefits and burdens of the process. It is thus imperative to consider these standards when dealing with Buy / Sell subcontracting.

Both IFRS books have incredible relevance and lay the regulatory groundwork for this subcontracting scenario as well as provide legal legitimacy to any accounting scenario as the standard accepted by most countries of the world. This section tries to only introduce the recurring concepts that are relevant to this case from the applicable books, however, it must be kept in mind that there is a huge number of standards that may be applicable in varying cases, which is why having a general grasp of the IFRS as a whole is important.

¹³ (IFRS Foundation, S. para. 5)

¹⁴ (IFRS Foundation, S. para. 39)

3.2 Transfer of Ownership and Repurchase Agreements

3.2.1 Transfer of Ownership

In the world of business, ownership transfer is a concept that involves the exchange of rights, responsibilities, and control over an asset or property from one party to another. This process greatly impacts the dynamics of relationships within a supply chain by determining when ownership of an asset or good changes hands. Typically this transfer is documented through the creation of an invoice, which serves as proof of the shift in ownership. Such strategic maneuvers affect the distribution of risk, responsibilities, and financial liabilities throughout the supply chain.

In the context of buy/sell subcontracting, transfer of ownership plays a particularly key role; because this process forms the basis of agreements between parties in the supply chain. At the beginning, it is confusing to decide whether the ownership passes to SP in a buy/sell scenario since there is a predetermined set of guidelines from the OEM that dictates the goods sold to SP, and it is also unclear whether there is a repurchase agreement between the parties. If so, according to IFRS 15, this transaction is to be considered as ‘forward’, and transfer of ownership to the SP does not occur.

Further along the process, the creation of an invoice marks a significant step for the subcontracting process. This formal document symbolizes the *completion of the ownership transfer* and also serves as a clear record of the transaction, providing a structured basis for further collaboration between the parties. Besides, as the SP holds the goods in its inventory and bears the holding costs for these goods, risk prima facie passes with the property.¹⁵ Therefore through the sale, ownership is transferred to the SP, which is independently responsible for the development, and sourcing of additional sub-suppliers alongside the production of the component and assumes the warranty and liability risks as well as inventory management for the goods transferred.¹⁶

¹⁵ (Arora, 2023)

¹⁶ (Mangold, 2019)

3.2.2 Advantages of Transfer of Ownership

The most significant advantage of transferring ownership lies in risk reduction, as the transfer involves shifting associated risks related to the goods. For OEMs transferring risks include those tied to holding materials, such as damage, loss, or obsolescence. This allows OEMs to mitigate risks during this production phase.

From the SPs perspective, one major advantage of ownership transfer is optimal resource utilization. Since SPs are involved in production due to their skills and expertise having control over materials enables utilization within the production process.

Furthermore, considering the perspective of the equipment manufacturer (OEM) this approach leads to a unique, superior and efficient final product.

3.2.3 Drawbacks of Transfer of Ownership

While the transfer of ownership brings benefits to both parties involved, it also presents disadvantages. Once the OEM relinquishes ownership of the goods, they become dependent on the service provider (SP), who becomes a player in the OEM's operations. Any shortcomings or financial and operational challenges faced by the SP during the processing phase could have an impact on the quality, timing, or delivery of the finished products back to the OEM.

Another drawback arises from issues in terms of accounting processes. The transfer of control can complicate matters. Place additional burdens on companies in terms of administrative and legal complexities. Additionally, as discussed in this paper, transferring control raises concerns about accounting standards. When revenue should be recognized under these circumstances. Such complexities make it challenging to comply with International Accounting Standards regulations such as IFRS, US GAAP and HGB.

3.2.4 Repurchase Agreements

Although there are similarities between buy/sell subcontracting processes and repurchase agreements, they are, for purposes that are to be demonstrated in a moment, fundamentally different.

To begin with, when it comes to a forward option for an asset the SP does not actually gain ownership or control over the goods. This is because the customers' ability to direct the use of and obtain most of the benefits from the asset is limited. As a result, the contract is treated as either a lease according to IFRS 16 or a financing arrangement. On the other hand, in a buy/sell subcontracting scenario, the OEM typically plays a direct role in development sourcing and demonstrates greater control compared to typical repurchase agreements. Additionally, in subcontracting processes, there is a shift in ownership from the customer to the SP. This shift is what primarily differentiates a repurchase agreement from a buy/sell subcontracting model.

There is another argument that arises regarding whether the assets repurchased from the subcontracting process are completely different from the provided material or are, in fact, substantially the same.¹⁷ To distinguish between these cases, analyzing and determining what "substantially the same" means is necessary.

ASC 470-40 Product Financing Arrangements, published by FASB which closely collaborates with IASB on determining accounting standards globally, dictate if an arrangement involving the sale of inventory and subsequent obligated repurchase of that inventory or an identical product can be categorized as a financing arrangement. However, when applying this metric to the Buy/Sell scenario, several challenges arise because the repurchased product must be substantially identical. This creates inconsistency because the service provider (SP) performs tasks and occasionally adds components, in the process altering the goods drastically in cases. Consequently, meeting this condition becomes complex and inconsistent.

Additionally, according to a staff paper published by IASB, it is agreed by the staff that for a transaction to be treated as a financing arrangement rather than a sale of the transferred asset, it should be required that it is "substantially the same." The staff also does not provide guidance on precisely defining what "substantially the same" means. After careful consideration, it has been concluded that the term "substantially the same" doesn't apply to how it is used. This means that the product has undergone changes and improvements. This distinction is crucial because it aligns with the specific

¹⁷ (IASB Staff, 2010)

goals and processes of the buy/sell subcontracting process. As the components evolve, this results in a product that has diverged significantly from the original assets or materials. It highlights the importance of interpreting this condition in a nuanced manner within our context.

3.3 Revenue Recognition

It is important to consider that revenue recognition can be quite complex and requires analysis in subcontracting situations, particularly in the buy/sell scenario. According to IFRS 15, revenue recognition revolves around delivering promised goods or services to customers while reflecting the expected consideration for the company. However, when a subcontractor buys components from an equipment manufacturer (OEM), this transaction does not typically fall under a customer contract as defined by IFRS 15.

As per Wiley's literature, IFRS 15 is only applicable when entering into a contract with a customer. In this context, a customer refers to a party engaging in a transaction for purchasing goods or services with other value as part of the business operations. This definition excludes transactions where both parties engage in activities or share risks and benefits without being considered clients. Therefore in this specific scenario of buy/sell subcontracting, the OEM does not recognize revenue from selling components to the subcontractor since they are not seen as customers in the sense.¹⁸

Moreover, this interpretation has implications on accounting practices for both OEMs and subcontractors. Equipment manufacturers (OEMs) face a challenge when recording financial activities in their records since the absence of revenue recognition in these transactions requires them to find ways to accurately represent these activities. Service providers (SPs) are also affected by this situation as it influences how they report their expenses and assets. The costs related to acquiring components can be expensed, depending on how those components are specifically used in the SP's operations.

Furthermore, this approach to recognizing revenue impacts how financial statements are presented and subsequently analyzed. For instance, an OEM may report sales figures but lower associated costs,

¹⁸ (Wiley Team, 2023, S. page 485)

which could potentially create a misleading portrayal of their financial well-being if not properly understood. In order to accurately grasp the financial condition of firms that are actively involved in subcontracting arrangements, these intricacies must be kept in mind.

It is equally important to consider variations across jurisdictions or accounting standards like US GAAP as they may have interpretations or applications of revenue recognition principles in similar scenarios. This array of accounting standards necessitates a well-informed approach toward reporting and analysis, particularly for multinational corporations where multiple accounting frameworks may come into play.

It is essential that standards of IFRS 15 should be interpreted and applied accurately because it ensures fair, transparent and reliable reporting. IFRS 15 highlights accurate accounting practices in acknowledging revenue under buy/ sell subcontracting circumstances. Therefore, this requires a thorough understanding of the transactional aspect as well as how the entities are intertwined.

3.4 Accounting for the Subject of the Subcontract

Accounting for the materials involved in the Buy-Sell scenario can be evaluated from two perspectives: the OEM and the SP. As discussed in the previous section, the "sale" of materials to the SP doesn't generate revenue for the OEM. However, once control is transferred to the SP after the purchase order is placed, and the materials are shipped, the OEM needs to remove these assets from its inventory. Even though a cash payment may not occur at this stage, the SP must pay the OEM. Consequently, an Accounts Receivable entry should be made by the OEM. According to the Japanese Subcontracting Act, in a Buy-Sell scenario, the SP should not make direct cash payments to the OEM during this process. This is to protect the liquidity of SPs, as most of them are SMEs compared to OEMs.¹⁹ The same transaction, from the SP's perspective, involves recognizing inventory and crediting Accounts Payable. Once the service is completed and the processed goods are shipped to the OEM following a purchase order, the OEM needs to recognize inventory, credit Accounts Payable, and close Accounts Receivable related to the previous transaction.

¹⁹ (Japan Fair Trade Commission)

Image-2: T-Balances

disclose all of the various risks they bear in their financial statements. These risks could be in the form of credit, liquidity, and market risk.

IAS 37 comes into play when uncertainties related to the subcontracting process arise. This standard addresses the need for provisions or disclosure of contingent liabilities in the event of potential delays or defaults by the SP. In addition to that, this standard highlights how these contingent liabilities, where the likelihood is not remote, should be disclosed, making sure that the stakeholders are aware of potential off-balance sheet risks.

Lastly, IAS 1 - Presentation of Financial Statements should be considered. “IAS 1 refers to financial statements as “a structured representation of the financial position and financial performance of an entity” and goes on to explain that the objective of financial statements is to provide information about an entity’s financial position, its financial performance, and its cash flows, which is then utilized by a wide spectrum of end users in making economic decisions. [...] All this information, and other information presented in the notes, helps users of financial statements to predict the entity’s future cash flows and their timing and certainty.”²⁰ The implications of purchase obligations under subcontracting agreements are vital to a company's financial health, affecting its liquidity and solvency. Firms must be careful when there is ambiguity and follow the path of transparency and prudence. For instance, a company should carry out a thorough analysis when evaluating the likelihood of a default or any other potential disputes under IAS 37. Firms should lean on comprehensive disclosures so that the stakeholders have a broad and clear understanding of the risks and commitments inherent to these transactions.

In conclusion, utilization of IFRS 7, IAS 1 and 37 shouldn't just be limited to the technical requirements. Firms should embrace the actual principles behind these standards. Using this approach, firms ensure a faithful representation of the financial well-being implications of the subcontracting agreements.

²⁰ (Wiley Team, 2023, S. page 44-45)

3.6 Contingency & Bankruptcy

3.6.1 Credit Risk

As with any future cash flow process, there is a significant credit risk within Buy/Sell subcontracting. It is not impossible that one party is not able to fulfill an obligation and default on it, and with the subcontracting process being heavily reliant on the commitments being fulfilled by both sides, one party's inability or unwillingness to fulfill its performance obligations damages both sides financially. This means a comprehensive guideline for fairly arbitrating such scenarios is required to prevent financial burden being unfairly put on the other party.

Credit risk exists for OEM and SP, i.e., the risk that the obligations will not be fulfilled. To place the subcontracting process on top of a secure, financially sound foundation, disclosures of credit risk involved and exposure to credit risk are necessitated.

3.6.2 IFRS Credit Risk Disclosure

IFRS 7 provides a set of guidelines as to how credit risk disclosures are to be prepared. These include the requirements to provide detailed information about the entity's credit risk management practices, quantitative and qualitative description of the potential credit losses' effect on the financial statements as well as information about the entity's credit risk exposure.²¹ While these are the recommended regulations, it should be kept in mind that depending on how much information is required to fulfill the objective of enabling the user of the financial statements to "(...) understand the effect of credit risk on the amount, timing and uncertainty of future cash flows."²²

The previously mentioned criteria for a complete credit loss disclosure, namely the risk management policies of the entity, qualitative & quantitative information, and exposure, are further elaborated upon in IFRS 7, which provides a widely applicable definition of these concepts to ensure they provide the necessary information. While it is essential to properly implement these disclosure practices, it is also true that these standards lie beyond the scope of this paper as they relate heavily

²¹ (IFRS Foundation, S. para. 35b)

²² (IFRS Foundation, S. para. 35b,d,c,e)

to financing arrangements in contrast to the mainly manufacturing-oriented process of Buy / Sell subcontracting.

3.6.3 Perspective of the OEM

From the perspective of the OEM, the timing of declaring bankruptcy divides the process into two different scenarios: The first occurs if the SP goes bankrupt before the sold material has been processed, and the second happens during the production process of the sold material. There is no scenario regarding defaults, which means the case where SP cannot make their payment because the netting principle is considered since the liquidity of the SP shouldn't be affected according to the Japanese Subcontracting Act.

In both scenarios, as all the firms consider, OEM is also aware that selling raw materials results in an entry in the Account Receivables. However, these receivables also come with a risk of unpaid amounts. That's why if SP is uncertain to fulfill its obligations, OEM should already present some portion of the receivables on the Allowances for the Doubtful Accounts. Then, either if the first scenario or the second scenario occurs, this allowance should be written off as bad debt.

3.6.4 Perspective of the SP

If the company that supplies materials, known as the OEM, goes bankrupt, it would create challenges for SP. One particular concern would be the materials that have already been delivered by the OEM. If the OEM is unable to buy those materials due to their insolvency, it could lead to problems for subcontractors who are owed money by the OEM. According to accounting principles like IFRS 9, subcontractors would have to reassess whether they can recover the amounts owed by the OEM. In this situation, subcontractors would need to recognize and calculate expected credit losses on these amounts based on factors such as terms and the OEM's inability to fulfill its obligations. Subcontractors may also need to set aside a provision for these expected credit losses, taking into account write-offs and an updated assessment of the default risk multiplier, including any increased country risk premium associated with the OEM's insolvency. This provision aims to minimize impact

and adjust how much subcontractors show in their statements for trade receivables or contract assets so that it more accurately reflects what they expect to recover.

3.7 Disclosure of Contractual Obligations: The Six-Step Disclosure Model

1. Nature and Purpose of the Liabilities

The first step of the six-step model consists of determining the nature and the goal of the entity's resulting liabilities from entering into a Buy/Sell subcontracting agreement. This is accomplished by first defining the participating entities who enter binding obligations following the requirements outlined in the agreement, after which the object of the agreement is highlighted. The dates determined within the agreement for procurement and delivery are also mentioned, alongside the purpose of the agreement itself, for the sake of clarity and transparency.

2. Risks and Contingencies

The risks involved in subcontracting often have a very tangible effect on the entity's financial standing, including the future cash flows, which significantly impact liquidity and the possibility of credit losses due to the non-fulfillment of obligations. As such, the second step of the model involves illustrating the risks that arise from the subcontracting process and how these can ultimately affect the firm's financials. This step refrains from focusing heavily on providing quantitative information and serves mainly to clarify the risks involved.

3. Quantitative Information

As the name implies, in this step, the quantitative information regarding the agreement is illustrated. These include:

- The amount that the OEM is obliged to pay to purchase the manufactured components and consideration for the materials initially sold to the SP.
- Relevant dates for the subcontracting agreement, such as the sale of materials or the purchase of finished components by the OEM, clearly depict the obligations involved.
- Expected credit losses arising from non-fulfillment.

- A proportional depiction of the contractual obligations from subcontracting to a firm's liquidity reserves paints a clearer picture of how fulfilling these obligations affects liquidity.

4. Qualitative Information

This section briefly explains how management conducts the supply chain processes, such as SP diversification, and manages the risk and liquidity of the firm by using metrics.

5. Explanation of Accounting Rules

All the steps of the model must be aligned with IFRS regulations, therefore in this section, it is briefly mentioned which IFRS policy applies.

6. Future Cash Flow Implications

While several obligations are incurred within the subcontracting agreement throughout the process, right from the initial transfer of materials, the respective cash flows occur at the very end, where a netted amount of cash is transferred between the parties. The final effect on liquidity can be observed in this step, representing the final cash transfer that concludes the subcontracting process.

4. Real World Examples and the Impact on Financial Statements

The importance of transparently disclosing subcontracting commitments and risks is evident while analyzing the financial statements of leading companies like Ferrari, renowned for its automotive prowess, and Nvidia, a multinational technology giant. As an automotive manufacturer, Ferrari already provides comprehensive details within its financial statements, presenting their purchase obligations and providing qualitative and quantitative insights. Conversely, Nvidia, operating across hardware manufacturing, artificial intelligence, and gaming sectors, heavily relies on subcontracting to ensure seamless service provision and avert supply deficits. This analysis digs into both companies' financial statements through the six-step disclosure model. This comparative assessment offers a nuanced understanding of how Ferrari and Nvidia communicate and manage subcontracting risks within their respective economic landscapes.

Company	NVIDIA	Ferrari N.V
Accounting Standards	US GAAP	IFRS
Nature and Purpose of the Liabilities	The firm states in its financial report that “We do not manufacture semiconductors used for our products. Instead, we utilize a fabless manufacturing strategy, whereby we employ world-class suppliers for all phases of the manufacturing process. (...) We utilize independent subcontractors, such as Amkor Technology, King Yuan Electronics Co., Ltd., Omni Logistics, LLC, Siliconware Precision Industries Company Ltd., and Wistron Corporation to perform assembly, testing, and packaging of most of our products and platforms. (...) We typically receive semiconductor products from our subcontractors, perform incoming quality assurance and configuration using test equipment purchased from industry-leading suppliers such as Advantest America Inc.”. By explicitly naming the entities involved, as well as the object of the subcontracting process, the first steps for a comprehensive illustration of the process are accomplished.	Ferrari discloses information about their subcontracting process under Notes: 29. Commitments section: “From time to time, in the ordinary course of business, the Group enters into various arrangements with key third party suppliers to establish strategic and technological advantages.” These arrangements generate unconditional purchase obligations as well as off-balance sheet liabilities. Even though Ferrari mentions these processes on their financial statements, according to the Six-Step model, it would be better to include contracted parties, the contract's object, and the arrangement's critical dates.
Risks and Contingencies	The following statement “We depend on third parties and their technology to manufacture, assemble, test, package or design our products, which reduces our control over product quantity and quality, manufacturing yields, development, enhancement and product delivery schedule and could harm our business.” can be seen as a disclosure of the risks inherent to the subcontracting process the firm is involved in, as it outlines the possible disruptions within the manufacturing process caused by the firm relinquishing its control over a portion of the supply chain.	Ferrari’s financial statement pays close attention to the risks arising from their businesses; for example they disclosed information about their practice against credit risk against the credits granted for the purchase of cars and car parts. However, these risks are not specifically for the subcontracting process. As discussed previously in 3.7, in this section of the Six-Step Model, Ferrari could have included more specific risks arising from their subcontracting process, such as credit risks, market risks, or liquidity risks.
Quantitative Information	While the firm has provided an extensive description of the amounts attributable to purchase commitments and distributed to the relevant due dates, this information remains lacking as to provide a clear picture	According to the model, a firm should be discussing the amount owed by the SP, owed to the SP, due dates, and expected credit loss from failure to commit to the contract. Ferrari already discloses the amount

	regarding the quantitative aspect of the subcontracting processes. Furthermore, it is stated that “We perform ongoing credit evaluations of our customers’ financial condition and maintain an allowance for potential credit losses. This allowance consists of an amount identified for specific customers and an amount based on overall estimated exposure.” asserting that credit risk and expected losses arising from credit loss are kept in mind. Despite this, the risk profile of subcontracting agreements are not provided in a quantitative manner that can be classified as disclosure or directly attributable to the agreement. The only piece of quantitative information we’re left with is ultimately the purchase obligations as mentioned initially and depicted in the table below, which by itself is insufficient to make a definite statement about the qualitative aspect of the process.	they owe to different SPs in a table format below. Although this is more transparent than the industry average regarding reporting standards, it can be improved further by sharing quantitative information on the trade receivables related to these specific arrangements. (Ferrari fs 2022 pg.337)
Qualitative Information	Nvidia heavily relies on external suppliers for the manufacturing of their products including semiconductors and more, with Buy / Sell subcontracting being but one example of such methods for inventory procurement. As a result, the financial effects of external circumstances in the business environment are of critical interest. The firm provides key characteristics of its supply and demand alongside prominent external changes in its financial report. While informative about the firm’s general financial standing, a complete implementation of the six step model would include more information about the firm’s subcontracting approach such as a comprehensive illustration of the firm’s SP network as well as how this is diversified and the effect of subcontracting on liquidity and the company’s risk profile in general.	Ferrari's strategic agreements with key suppliers and sponsors demonstrate a sophisticated approach to supply chain management; however, they don’t disclose any other important information on how they mitigate risks during the subcontracting process. The strategies they use against financial dangers, in general, are presented under the Qualitative and Quantitative Information on Financial Risks section, which is again not specific to this process.

Explanation of Accounting Rules	The US-GAAP (Generally Accepted Accounting Principles) is the main standard which the firm conforms to, as the financial report in question is a Form 10-K filed to the SEC.	It can be seen on the FS of the firm that they adhere to the principles outlined in IFRS. This compliance ensures that Ferrari's reports are transparent, reliable, and comparable to other firms on a global scale. However, the report doesn't emphasize the accounting standards followed for the subcontracting agreements. As this case is quite ambiguous in the original IFRS text, the company should also mention how they interpret the relevant provisions.
Future Cash Flow Implications	The future cash flow implications of the firm's subcontracting processes are elaborated upon in the purchase obligations section of the notes to the financial report. The dates as well as the aggregated size of the obligations due are provided in the table which can be found under step three, however a clear cut display of cash in- or outflows are lacking. Netted cash flows resulting from the subcontracting agreement are one of the central transactions within subcontracting, therefore must be illustrated in the six-step model.	Since in Ferrari's FS, the subcontracting process, subcontracting type, payment conditions, or the amounts are not presented precisely, it is impossible to interpret this process in the scope of cash flows. The only implications on the effect of future cash flows can be seen on the general table with due dates. If Ferrari had been more specific about the contracts, the subcontracting results on future cash flow could have been discussed in depth.

Total future unconditional purchase commitments as of January 30, 2022, are as follows:

	Commitments (In millions)	
Fiscal Year:		
2023	\$	9,302
2024		765
2025		201
2026		28
Total	\$	10,296

Image-3: Purchase Commitments as Outlined in the FS of NVIDIA²³

At December 31, 2022					
(€ thousand)	Due within one year	Due between one and three years	Due between three and five years	Due beyond five years	Total
Minimum purchase obligations	54,964	12,445	3,311	—	70,720

Image-4: Purchase Commitments as Outlined in the FS of Ferrari N.V.²⁴

²³ (NVIDIA, 2022)

²⁴ (Ferrari N.V, 2022)

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